Logarithmic and Exponential Functions

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

[Exponential, Logarithmic & Logistic Dynamics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

2 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve the logarithmic equation $\log_3(x + 6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

3 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

A radioactive substance follows the decay model $M(t) = M_0 e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant k and the substance's half-life in days to the nearest tenth.

4 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve the logarithmic equation $\log_2(x + 3) + \log_2(x - 3) = 4$ for all real solutions x. State any extraneous roots.

5 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

If $f(x) = 4x - 3$ and $g(x) = x^2 + 1$, find (a) $gf(2)$ and (b) an expression for $g^{-1}(x)$ for $x \geq 1$.

6 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

[Probability & Statistics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

7 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

[Transformations & Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

[Unit Circle & Trigonometric Identities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

9 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

[Conic Sections & Analytic Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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10 **[Intermediate]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

[Polar Coordinates, Complex Numbers & De Moivre's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

11 **[Intermediate]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

[Sequences, Series, Induction & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

12 **[Intermediate]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

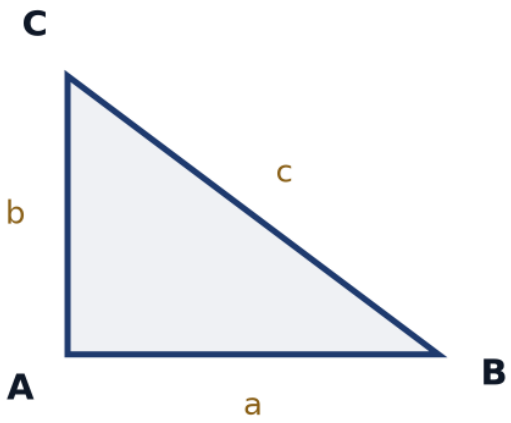
[Trigonometry & Triangle Laws · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

13 **[Intermediate]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

[Vectors, Lines & Parametric Kinematics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

14 **[Intermediate]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.

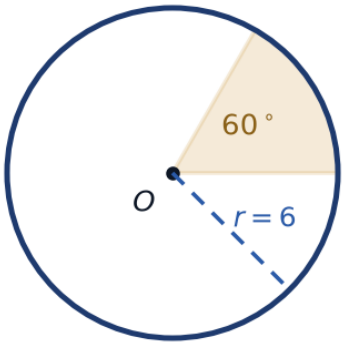


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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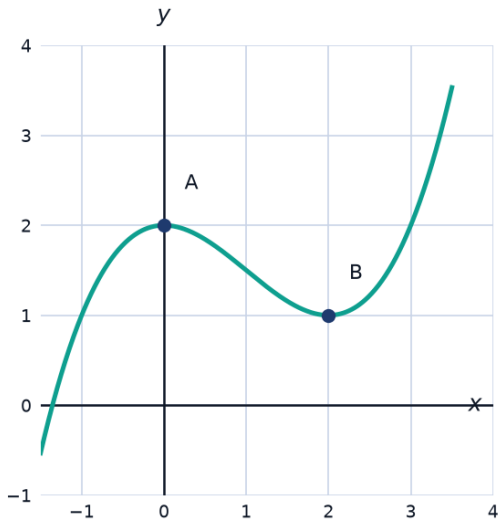
15 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.



16 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

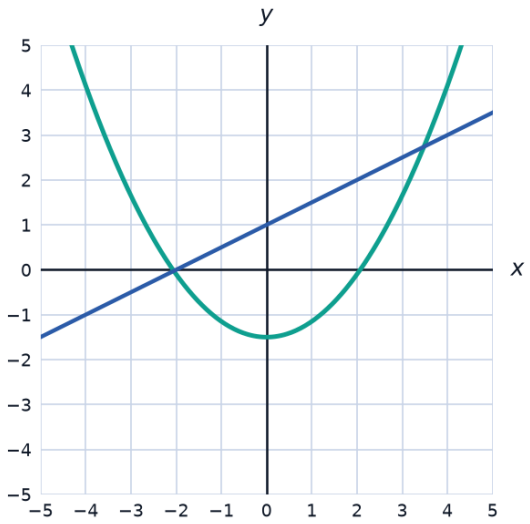
Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.



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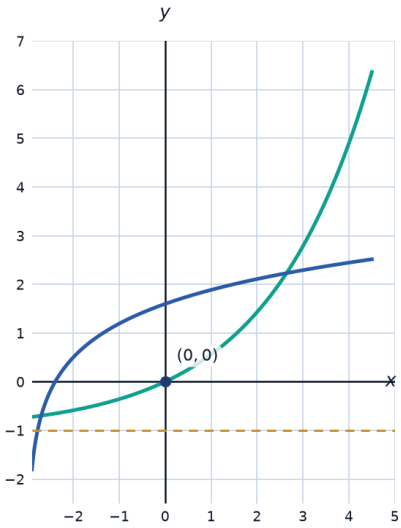
17 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD 6*cmandtheradiusisUSD*3 cm, find the angle θ in radians.



18 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.



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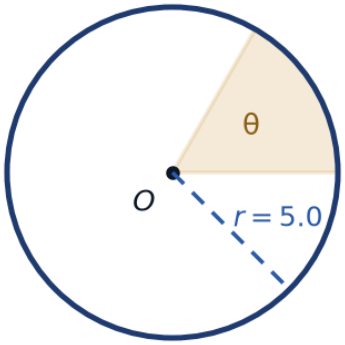
19 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).



20 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve $7^x = 100$ for x, expressed using logarithms.

a. $x = \frac{100}{7}$

b. $x = 100 - 7$

c. $x = \log_7(100) = \frac{\ln 100}{\ln 7}$

d. $x = \log_{100}(7)$

22 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

A population of 500 grows at 5% per year, modeled by $P(t) = 500(1 + 0.05)^t$. Approximately how many years until the population reaches $1.5\times$ its initial size (i.e. $P = 1.5 \cdot 500$)?

a. 8.3 years

b. 10.8 years

c. 6.3 years

d. 10.3 years

23 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Write $5^4 = 625$ in logarithmic form.

a. $\log_{625}(5) = 4$

b. $\log_5(4) = 625$

c. $\log_4(5) = 625$

d. $\log_5(625) = 4$

24 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

b. $\log(x) \cdot \log(y)$

c. $\log x + \log y$

d. $\log x - \log y$

25 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms.

a. $\log(x) \cdot \log(y)$

b. $2\log x - 3\log y$

c. $2\log x + 3\log y$

d. $\frac{\log x}{\log y}$

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26 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Evaluate the definite integral: $\int_0^1 (2x+1)^3\,dx$.

a. $\frac{80}{2}$

b. $\frac{84}{8}$

c. $\frac{80}{4}$

d. $\frac{80}{8}$

27 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Find the number of distinct permutations of all letters in STATISTICS if all 3 'S's must stay together as a single block.

a. 3360

b. 1680

c. 6720

d. 50400

28 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Find the number of distinct permutations of all letters in COMMITTEE if the 2 'M's must stay together.

a. 10080

b. 20160

c. 45360

d. 5040

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

A committee of 4 people is to be chosen from 5 men and 4 women. How many committees consist of exactly 2 men and 2 women?

a. 60

b. 120

c. 55

d. 36

30 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Find the number of distinct permutations of all letters in ADDITION if both 'D's must stay together.

a. 10080

b. 5040

c. 2520

d. 1260

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31 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Find the number of distinct permutations of all letters in ARRANGE if the 2 'R's must be together.

a. 180

b. 360

c. 720

d. 1260

32 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Find the number of distinct permutations of all letters in SUCCESS if all 3 'S's must stay together.

a. 30

b. 420

c. 120

d. 60

33 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Find the number of distinct permutations of all letters in PARALLEL if all 3 'L's must stay together as a single block.

a. 180

b. 720

c. 360

d. 3360

34 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of the parameter a.

a. 1

b. 2

c. 3

d. 4

35 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

A population of bacteria, initially P_0 cells, grows according to the logistic model $P(t) = \frac{L}{1 + Ae^{-kt}}$, where t is in hours, L is the carrying capacity, and A and k are positive constants. At $t = 0$, the population is $P(0) = \frac{L}{3}$. After 2 hours, the population has doubled to $P(2) = \frac{2L}{3}$. Find the exact value of e^{2k} in terms of L and P_0.

a. $\frac{9}{4}$

b. $\frac{4}{9}$

c. 3

d. $\frac{3}{2}$

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36 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Consider the exponential growth model $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t and P_0 is the initial population. If the initial population is $P_0 = 500$ and after 10 years the population is $P(10) = 1200$, find the annual growth rate k. Round your answer to four decimal places.

a. 0.0588

b. 0.0875

c. 0.1200

d. 0.0434

37 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Consider the exponential growth model described by the function $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

a. 0.347

b. 0.416

c. 0.301

d. 0.477

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

Consider the exponential function $f(x) = 2^x$. If $f(a) = 8$, what is the value of a?

a. 1

b. 2

c. 3

d. 4

39 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Consider the exponential function $f(x) = 2^x$. What is the value of $f(3)$?

a. 6

b. 8

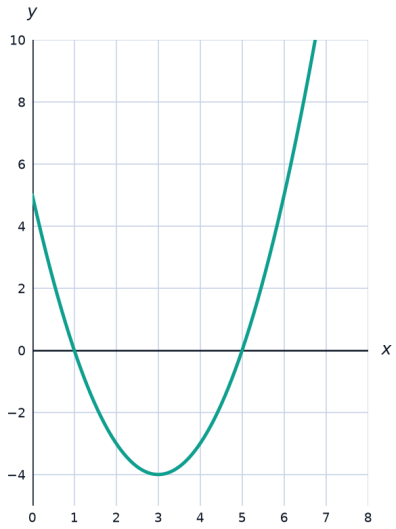
c. 9

d. 5

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40 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

A function is defined by $f(x) = x^2 - 6x + 5$ for $x \geq 3$. \\ The graph of $y = f(x)$ is a parabola. A straight line with equation $y = k$ intersects the graph of $y = f(x)$ at exactly one point. \\ Find the possible value(s) of k.



a. $k = -4$

b. $k = 5$

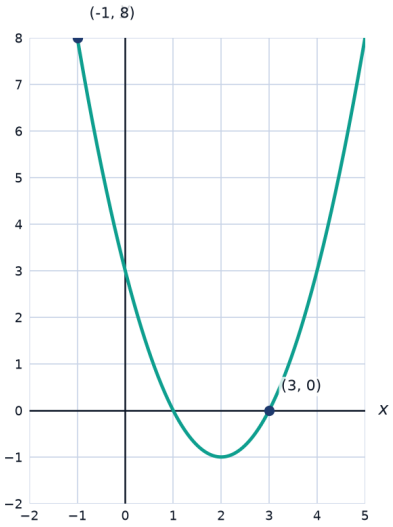
c. $k = -4$ or $k = 5$

d. $k \geq 5$

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41 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

The diagram shows the graph of the quadratic function $f(x) = x^2 + ax + b$. The graph passes through the points $(-1, 8)$ and $(3, 0)$. Find the values of a and b.



a. $a = -2, b = 3$

b. $a = 2, b = -3$

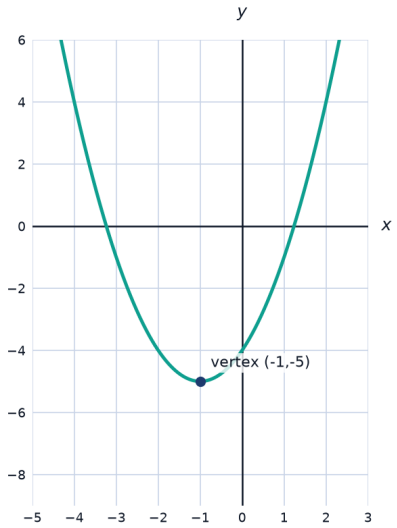
c. $a = -4, b = 1$

d. $a = 4, b = -1$

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42 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

The graph of $g(x) = f(x + 1) - 5$ is obtained from $f(x)$ by:



a. Shifting 1 units left and 5 units down

b. Shifting 1 units right and 5 units down

c. Shifting 1 units right and 5 units up

d. Reflecting over the x-axis

43 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

The function f is defined by $f(x) = x^2 - 6x + 13 \quad \text{for } x \geq 3$. Determine the complete set of real values of the constant k for which the equation $f^{-1}(x) = |2x - k|$ has exactly one real solution.

a. $k = \frac{39}{8}$ or $5 < k < 11$

b. $k = \frac{39}{8}$ or $5 \leq k < 11$

c. $\frac{39}{8} < k \leq 5$ or $k \geq 11$

d. $k = \frac{39}{8}$ only

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44 **[Challenging]** • **Target: Logarithmic & Exponential Functions Practice Exercises** • **Cross-Lesson: 7.01 Differentiation and Rates of Change**

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. Limit at a is 3, $\lim_{x \to b^-} f(x) = -\infty$

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

45 **[Challenging]** • **Target: Logarithmic & Exponential Functions Practice Exercises** • **Cross-Lesson: 8.01 Integration and Kinematics**

The function f is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k,$ where k is a constant. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - 4|.$ Given that $f^{-1}(4) = 2$, find the value of k and all real solutions to the equation $f^{-1}(x) = g(x)$.

a. $k = -2;\ x = \frac{5}{2},\ x = 7$

b. $k = -2; \; x = 7$

c. $k = 2;\ x = \frac{7}{2},\ x = 5$

d. $k = -2;\ x = -\frac{5}{2},\ x = 7$

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